UNC Charlotte

ECGR 3120 Electromagnetics

Magnetically Steered Cardiac Ablation

Milestone 3: Coil Power, Heating, and Mechanical Forces

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Submission date: _____

Input status: The coil geometry and illustrative ampere-turn product are inherited from the Milestone 1 example. Current, turns, wire radius, cooling properties, and design limits in this report are explicitly illustrative because official values were not in the workspace.

Abstract

This report estimates resistance, Joule heating, current density, free convection temperature rise, and mutual attractive force for a pair of navigation coils. The example uses the existing handout coil geometry and Milestone 1's explicitly illustrative ampere-turn product, along with clearly labeled example winding and cooling assumptions. A circular-loop force calculation and analytical checks are used to verify scaling, direction, and convergence. Since no official per-turn current, turn count, wire construction, or cooling parameters were present, the numerical design results are illustrative rather than claims of official baseline verification.

Contents

1 Objective and input status **1**

2 Governing equations **1**

3 Electrical and thermal calculation **2**

4 Mechanical force calculation **2**

5 Verification **3**

6 Figures **4**

7 Engineering recommendation **6**

8 Limitations **7**

9 How I Would Explain Milestone 3 to My Professor **7**

1 Objective and input status

Milestone 3 connects the navigation coils' electrical drive to power dissipation, heating, and mechanical support loads. The handout reference values are approximately 1.45 kW dissipated per coil, 690 N attraction, and a free-convection temperature rise greater than 250 K. They are comparisons, not automatically verified results.

The available project files specify coil radius $R = 0.35\,\mathrm{m}$ and coil spacing $s = 0.35\,\mathrm{m}$, and contain a Milestone 1 example ampere-turn product $NI = 31\,139.611$ A turns per coil derived from an illustrative 80 mT field. They do not specify official coil current, number of turns, wire dimensions, copper grade, cooling coefficient, winding-pack surface, or allowable temperature rise.

For a reproducible example, this report uses $N = 1000$ turns and derives $I = NI/N = 31.1396\,\mathrm{A}$ from that illustrative product. It assumes 4 mm wire radius, $\rho = 1.724 \times 10^{-8}\,\Omega\,\mathrm{m}$ (approximate copper resistivity near 20 °C), natural-convection coefficient $h = 5\,\mathrm{W\,m^{-2}\,K^{-1}}$, and winding fill fraction 0.65. This resistivity is equivalent to conductivity $\sigma = 1/\rho \approx 5.8005 \times 10^7\,\mathrm{S\,m^{-1}}$; the script exposes resistivity as the configurable material parameter. The 1000 W-per-coil power budget and 40 K maximum rise are illustrative design targets, not professor-provided limits. The handout's $4\,\mathrm{A\,mm^{-2}}$ current density is treated as a design reference, not a universal safe limit.

2 Governing equations

For one series winding of N circular turns, each of mean radius R, the total conductor length is

$$L = N(2\pi R).$$

For copper cross-sectional area A and resistivity ρ, resistance and Joule heating per coil are

$$R_{\mathrm{coil}} = \rho\frac{L}{A}, \qquad P_{\mathrm{coil}} = I^2 R_{\mathrm{coil}}, \qquad J = \frac{I}{A}.$$

Here I is the actual current in each series turn. NI sets the magnetizing field; it must not be substituted for I in I^2R.

The first-order lumped free-convection estimate is

$$\Delta T = \frac{P}{hA_{\mathrm{exposed}}}.$$

It assumes a uniform coil temperature, fixed electrical power and convection coefficient, and convection as the only heat-loss mechanism. A square-section toroidal winding pack with radial build w, axial build h_p, and mean radius R is approximated by

$$A_{\mathrm{exposed}} = 4\pi R(w + h_p).$$

The example pack cross-section is inferred from total copper area and the assumed packing fraction; it is an estimated external surface, not a measured cooling area.

For mechanical force, the field from the upper circular coil is evaluated at the lower coil's segment midpoints using the discretized Biot–Savart law. The complete lower winding is integrated as

$$\mathbf{F} = (NI)\sum_j \Delta\mathbf{l}_j \times \mathbf{B}_j.$$

The same current-loop geometry is used for both coils. The full circular geometry is necessary to calculate the mutual axial force; the straight parallel-conductor formula is used only in its separate reference check.

3 Electrical and thermal calculation

The example uses a 4 mm conductor radius, so the copper area is $50.2655\,\mathrm{mm}^2$. With 1000 turns at 0.35 m mean radius, the wire length is 2199.115 m per coil. The resulting resistance is $0.754\,250\,\Omega$, and actual per-turn current gives

$$P_{\mathrm{coil}} = (31.1396\,\mathrm{A})^2(0.754\,250\,\Omega) = 731.378\,\mathrm{W}.$$

The pair therefore dissipates 1462.755 W in total in this illustrative example. This is not the handout's 1.45 kW per-coil result.

Current density is

$$J = \frac{31.1396\,\mathrm{A}}{50.2655\,\mathrm{mm}^2} = 0.619\,503\,\mathrm{A\,mm^{-2}}.$$

For the handout's $4\,\mathrm{A\,mm^{-2}}$ design reference, the example current requires a wire radius of at least 1.574 mm. The separate illustrative 1000 W-per-coil power budget requires radius of at least 3.421 mm at fixed current and turns. Thus, the selected 4 mm radius satisfies both selected example criteria.

The assumed square winding-pack side is 0.278 085 m, yielding estimated exposed area $2.446\,167\,\mathrm{m}^2$. With $h = 5\,\mathrm{W\,m^{-2}\,K^{-1}}$, the lumped rise is 59.798 K. Against the illustrative 40 K target, this passive-convection model fails. At the assumed h, meeting that rise would require $3.656\,888\,\mathrm{m}^2$ exposed surface per coil. Alternatively, at the estimated current surface, the required convection coefficient would be $7.475\,\mathrm{W\,m^{-2}\,K^{-1}}$. Neither the target nor the assumed surface and h are official specifications, so this is a preliminary sizing result, not a final cooling-system selection.

4 Mechanical force calculation

The computed 512-segment force vector on the lower coil is

$$\mathbf{F} = [-5.91 \times 10^{-14},\ -1.09 \times 10^{-13},\ 696.575229]\ \mathrm{N}.$$

The negligible transverse components are numerical roundoff; positive z-force points toward the upper coil, so the force is attractive. The force magnitude is 696.575 N, which differs from the handout's approximately 690 N by 0.953% for this illustrative ampere-turn setting. That proximity does not validate the missing official current/turn configuration.

5 Verification

Table 1: Analytical and numerical checks

Check	Result	Status
Ulaby Exercise 4-14, 50 m copper wire, 2 cm radius	6.859578×10^{-4} Ω, 0.586% error	PASS
Parallel conductors, 1 A each and 1 m apart	2.000000×10^{-7} N/m	PASS
Resistance with doubled test area	0.017240 to 0.008620 Ω	PASS
Joule heating at 1, 2, 3 A and fixed R	0.75425, 3.01700, 6.78825 W	PASS
Current density with doubled area	0.619503 to 0.309751 A/mmš	PASS
Coil force at $0.5I$, I, $2I$	174.139, 696.556, 2786.225 N; ratios 0.25, 1, 4	PASS
Force direction	$F_z = +696.575$ N; transverse 1.237×10^{-13} N	PASS
512 vs. 1024 segment axial force	0.000677% difference	PASS

For Ulaby Exercise 4-14, the calculation uses $\rho = 1.724 \times 10^{-8}\,\Omega\,\mathrm{m}$, giving $R = 6.859\,578 \times 10^{-4}\,\Omega$, within 0.586% of the approximate $6.9 \times 10^{-4}\,\Omega$ reference. A local 2% rounding tolerance is used because the reference is approximate, not a professor acceptance threshold. The parallel-wire result uses $F/L = \mu_0 I_1 I_2/(2\pi d)$. No official `tests/test_m3_thermal.py` was present, so acceptance tests were unavailable and were not claimed as passed.

6 Figures

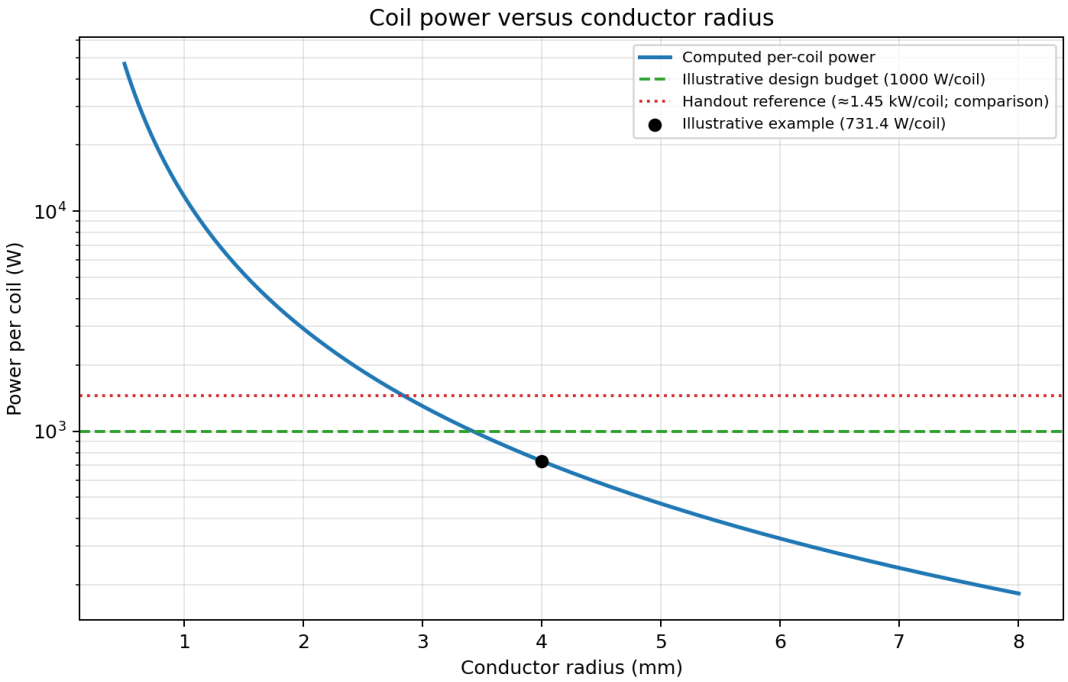


Figure 1: Computed power per coil versus conductor radius at fixed current and turn count. The 1 kW-per-coil line is an illustrative design budget; the 1.45 kW-per-coil line is a separate handout reference only. The marker is the illustrative 4 mm wire example.

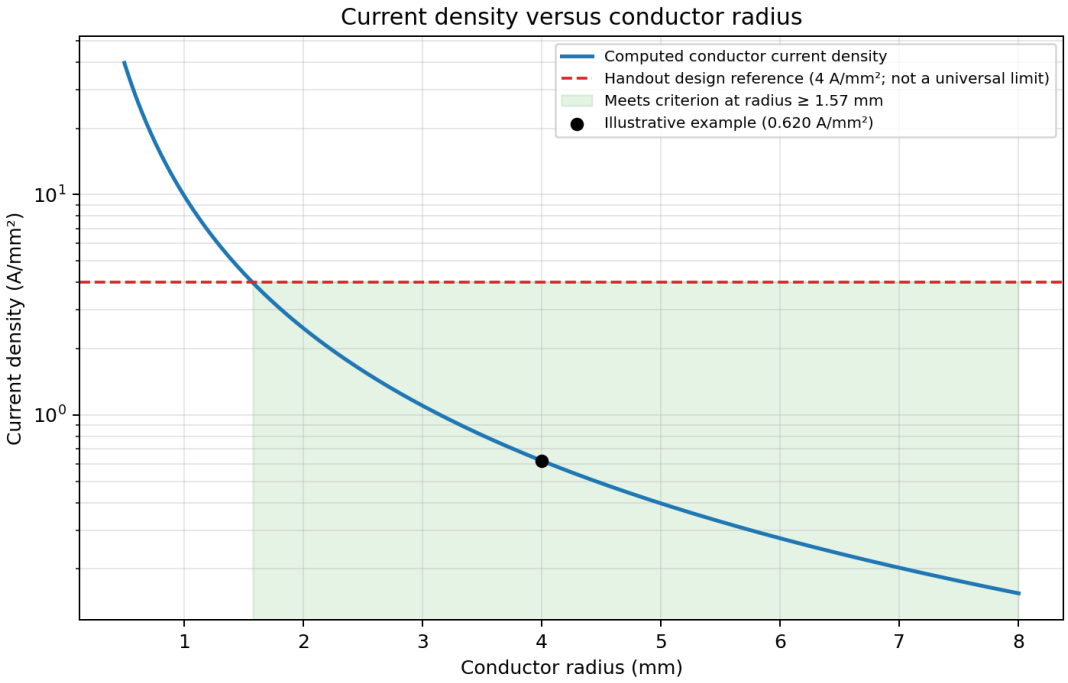


Figure 2: Computed current density versus wire radius. The 4 A/mmš line is the handout's design reference, not a universal current-density limit. Green shading indicates sizes satisfying that reference for the illustrative current.

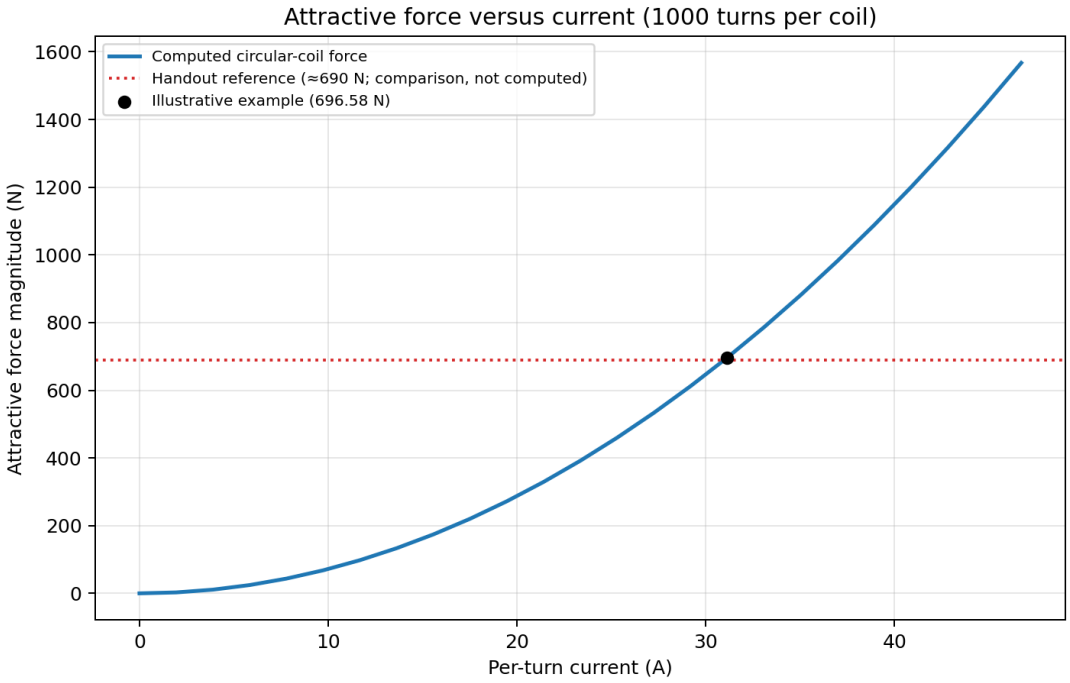


Figure 3: Attractive force calculated by full circular-coil integration versus current for fixed geometry and 1000 turns per coil. The approximately 690 N handout value is displayed only as a comparison.

7 Engineering recommendation

At fixed current and turns, using a larger conductor reduces resistance, power, and current density. In this example, the 4 mm radius satisfies both the illustrative 1 kW-per-coil power budget and the 4 A/mmš current-density reference. Actual conductor selection still depends on the official current, wire insulation, winding arrangement, and verified thermal limits.

The model's 59.798 K free-convection rise exceeds the example 40 K target. Under these assumptions, passive convection is not enough to meet that target. Increase exposed surface or convection, or consider a cooling approach that can reject the calculated 731.378 W per coil. The calculated area and convection coefficient requirements are preliminary: measure or establish the actual winding pack, airflow, thermal contact, and allowable winding temperature before choosing cooling hardware.

The two coils attract with approximately 697 N each under the illustrative drive. Coil supports and mounts should be designed against the maximum force recalculated from confirmed current, turns, and geometry, with the project's specified structural safety factor and any transient loads. The 690 N handout reference is not by itself a structural design allowable.

8 Limitations

The resistance model uses a single mean-radius length for every turn and constant copper resistivity. It omits wire leads, joints, skin and proximity effects, and temperature feedback. Thermal estimates use an assumed square pack, packing factor, external area, and lumped free-convection coefficient; they omit radiation, conduction, forced flow, internal hot spots, and coil insulation limits. The force uses ideal coincident circular turns and omits finite winding cross section, leads, structural flexibility, misalignment, and dynamic loads. The handout reference values need the official baseline inputs for direct reproduction.

9 How I Would Explain Milestone 3 to My Professor

I first used the actual current in each turn and the total series-wire resistance to calculate I^2R; the ampere-turn product belongs in the magnetic-field calculation, not in place of current in the heating equation. Then I checked wire resistance and current-density trends against area and verified the expected square-current scaling. For mechanical loading, I integrated the field and force around complete circular coils and checked that the lower coil is pulled toward the upper one and that the result converges as segments are refined. The default values are explicitly illustrative because the official current, turns, wire, and cooling inputs were not provided. The resulting power, temperature rise, and support force are therefore a transparent example, not a claim that the handout's references have been independently reproduced.